Human Reproduction

The Male Reproductive System

1. The seminiferous tubules are lined by two types of cells, that is,
1) Sertoli cells and Leydig cells
2) Sertoli cells and Nurse cells
3) Spermatogonial cells and Sertoli cells
4) Spermatogonial cells and Leydig cells

2. Which of the following is an incorrect statement?
1) Sperms are produced by cells of seminiferous tubules.
2) Sugar that sperms use for energy is called sucrose.
3) Testis is the primary male sex organ.
4) Testes are present outside the abdominal cavity in scrotum.

3. Read the following statements and choose the correct option.
1) The testes are situated inside the abdominal cavity.
2) In adults, each testis is oval in shape.
3) Width of each testis is about 3 to 4 cm:
4) Length of testis is about 10 to 12 cm

4. Each testis has about X compartments called testicular lobules. Each lobule contains Y highly folded seminiferous tubules in which spermatogenesis occurs.
Choose the option which correctly fills blanks X and Y.
1) X: 100; Y: 3-5
2) X: 250; Y: 1-3
3) X: 400; Y: 1-2
4) X: 200 Y: 7-10

5. Read the following statements:
A. Testis is divided into compartments called testicular lobules.
B. In males, the reproductive and excretory systems share urethra.
C. Bulbourethral glands and prostate glands are paired in the male reproductive system.
D. Vasa efferentia are ductules leading from vas deferens to urethra.
Which of the statements are incorrect?
1) (A) and (B)
2) (B) and (C)
3) (C) and (D)
4) (A) and (D)

6. The secretions of seminal plasma are
1) Rich in fructose and calcium but don't have any enzymes
2) Rich in fructose and calcium and have certain enzymes
3) Rich in glucose and calcium
4) Rich in glucose but lack calcium

7. In humans, vasa efferentia leaves the testis and opens into epididymis located along the
1) Anterior surface of testis
2) Lower surface of testis
3) Posterior surface of testis
4) Superior surface of testis

8. Read the following statements and choose the incorrect option.
1) The testis is covered by a loose covering.

2) The scrotum helps in maintaining the low temperature of the testes.
3) Each lobule contains one to three highly coiled seminiferous tubules.
4) Penis is made up of special tissue that helps in erection of the penis to facilitate insemination.

9. Which one of the following is not a male accessory gland?
1) Seminal vesicle
2) Ampulla
3) Prostate gland
4) Bulbourethral gland

10. Which of the following statements is incorrect?
1) The secretions of bulbourethral glands help in lubrication of penis.
2) The testes are situated outside the abdominal cavity within a pouch called epididymis.
3) Secretions of epididymis, vas deferens, seminal vesicle, and prostate are essential for maturity and motility of sperms.
4) In testis, the regions outside the seminiferous tubules called interstitial spaces contains small blood vessels and interstitial cells or Leydig cells.

11. The vas deferens receives duct from the seminal vesicle and opens into urethra as
1) Epididymis
2) Ejaculatory duct
3) Efferent ductule
4) Ureter

12. Which of the following is an incorrect statement with respect to the male copulatory organ?
1) The penis is the male organ of copulation.
2) It is made up of a special tissue that helps in erection of the penis to facilitate insemination.
3) The enlarged end of penis called the glans penis is covered by a loose fold of skin called foreskin.
4) None of the above

13. Which of the following condition is observed if the temperature of scrotum rises by 2°C to 3°C?
1) It does not have any effect on fertility.
2) The testes are pulled into abdominal cavity, and scrotum becomes empty.
3) Spermatogenesis is inhibited, and the fertility reduced.
4) None of the above

14. The secretions from which of the following will not be seen in the semen if the vas deferens are cut and tied up?
1) Prostate gland
2) Seminiferous tubules
3) Bulbourethral gland
4) Cowper's gland

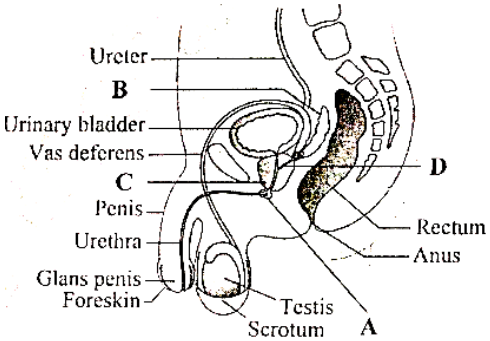
15. In males, the urethra originates from the urinary bladder and extends through the penis to its external opening called
1) Perineum
2) Urethral meatus
3) Ostium
4) External os

16. Which of the following is observed if one of the vas deferens is cut and ligated?
1) Semen does not contain any sperms
2) Sterilization would still be successful
3) Sterilization would not be successful
4) It reduces desire for sexual intercourse

17. The mucus secretion of which gland lubricates the penis during sexual intercourse?
1) Prostate gland
2) Bulbourethral gland
3) Seminal vesicle
4) All of the above

18. Urethral meatus refers to the
1) Urinogenital duct
2) Opening of vas deferens into urethra
3) External opening of the urinogenital duct
4) Muscles surrounding the urinogenital duct

19. Given here is a diagrammatic sectional view of male pelvis showing the reproductive system. Identify the labelings A, B, C, and D.



1) A-Bulbourethral gland; B-Seminal vesicle; C-Prostate gland; D-Ejaculatory duct
2) A-Seminal vesicle; B-Prostate gland; C-Bulbourethral gland; D-Ejaculatory duct
3) A-Prostate gland; B-Seminal vesicle; C-Bulbourethral gland; D-Vas deferens
4) A-Bulbourethral gland; B-Seminal vesicle C-Prostate gland; D-Vas deferens

The Female Reproductive System

20. Which of the following is not included in the external genitalia of females?
1) Labia majora
2) Labia minora
3) Cervix
4) Clitoris

21. Find the incorrect match.
1) Myometrium: Smooth muscle layer that contracts during parturition
2) Vagina: Lined internally by glands that are shed during menstruation
3) Isthmus: Last part of oviduct
4) Cervix and vagina: Birth canal

22. In the female external genitalia, the structure homologous with penis of male is
1) Vagina
2) Labia majora
3) Uterus
4) Clitoris

23. The layer of uterus that undergoes cyclic changes during menstrual cycle is
1) Endometrium
2) Myometrium
3) Perimetrium
4) Mesometrium

24. Identify the odd one from the following:
1) Labia minora
2) Fimbriae
3) Infundibulum
4) Isthmus

25. Match the column I with column II, and choose the correct option.

Column I		Column II	
A	Finger like structure	1	Infundibulum
B	Inverted pear like structure	2	Testes
C	Funnel-shaped structure	3	Fimbriae
D	Oval shaped structure	4	Uterus

1) a-1, b-4, c-3, d-2
2) a-3, b-2, c-1, d-4
3) a-4, b-3, c-2, d-1
4) a-3, b-4, c-1, d-2

26. Choose the correct sequence of ducts of mammary glands from areolae towards alveoli.
1) Mammary duct→Mammary ampulla→Lactiferous duct →Mammary lobe
2) Lactiferous duct→Mammary ampulla→Mammaryduct →Mammary lobe
3) Mammary lobes→Lacteriferous duct→Mammary ampulla→Mammary duct
4) Lactiferous duct→Mammary lobe→Mammary duct →Mammary ampulla

27. Consider the following three statements related to mammary glands, and select which ones are true (T) and which ones are false (F).

A: Mammary glands are paired structures with glandular tissues and variable amounts of fat.

B: The glandular tissue of each breast is divided into 250 to 300 mammary lobes containing clusters of cells called alveoli.

C: The alveoli opens into mammary ducts and these ducts from each lobe unite to form tubules.
1) A-T; B-F; C-T
2) A-T; B-F; C-F
3) A-F; B-T; C-F
4) A-T; B-T; C-F

28. All of the following statements about mammary glands are correct except one. Which one is wrong?
1) A functional mammary gland is characteristic of all female mammals
2) The glandular tissue of each breast is divided into 15 to 20 mammary lobes containing clusters of cells called alveoli.
3) Milk is sucked out through lactiferous ducts.
4) Mammary glands act as both exocrine and endocrine glands.

29. The part of fallopian tube with finger-like projections called fimbriae is
1) Ampulla
2) Isthmus
3) Infundibulum
4) Corpus

30. Identify the incorrect statement.
1) The shape of uterus is like an inverted pear.
2) Primary sex organs produce gametes and sex hormones.
3) Clitoris is a tiny finger-like structure that lies at the upper junction of two labia minora above the urethral opening.
4) Epididymis is an intermediate structure between rete testis and vasa efferentia.

31. How many of the following structures are paired in human?

Ovary, Oviduct, Prostate gland, Uterus, Cervix, Vagina, Hymen, Bulbourethral gland, Seminal vesicle, Penis

1) Four
2) Five
3) Six
4) Seven

32. Match the options given in Column I with those of Column II.

Column I		Column II	
A	Uterus	(i)	Primary sex organ
B	Clitoris	(ii)	Fimbriae
C	Ovary	(iii)	Womb
D	Infundibulum	(iv)	External genitalia

1) A-i; B-iii; C-ii; D-iv
2) A-iii; B-iv; C-i; D-ii
3) A-ii; B-i; C-iii; D-iv
4) A-iv; B-iii; C-ii; D-i

33. Find the option that contains odd one out related to their number in the human reproductive system.

1) Vagina, clitoris, penis, prostate
2) Ovaries, mammary gland, labia minora
3) Testes, ovaries, vas deferens
4) Prostate, vagina, bulbourethral gland

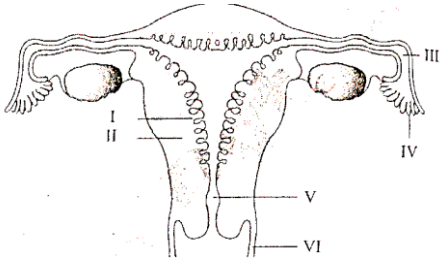
34. Which of the following structure is not included among female accessory ducts?

1) Uterus
2) Vagina
3) Fallopian tubes
4) Labia majora

35. Which of the following is an incorrect statement with respect to uterus?

1) The shape of uterus is like an inverted pear.
2) It is supported by ligaments attached to the pelvic wall.
3) Uterus opens into vagina through a narrow cervix.
4) None of the above.

36. Choose the correct option with respect to the labelling of the diagram given.



1) I-Endometrium; III-Isthmus, V-Uterus
2) II-Perimetrium; I-Myometrium, IV-Ampulla
3) I-Endometrium; V-Cervical canal, IV-Fimbriae
4) III-Ampulla; V-Uterus, VI-Cervical canal

37. The opening of vagina is covered partially by a membrane called

1) Labia minora
2) Hymen
3) Clitoris

4) Labia majora

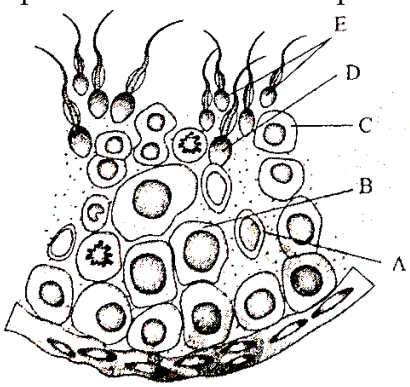
Gametogenesis and Menstrual Cycle

38. Which of the following hormones controls the process of spermiogenesis?
1) Luteinizing hormone
2) Follicle-stimulating hormone
3) Interstitial cell stimulating hormone
4) Gonadotropin-releasing hormone

39. Which of the following organelle is present in the middle piece of human sperm?
1) Nucleus
2) Acrosome
3) Golgi complex
4) Mitochondria

40. The immature male germ cell undergo division to produce sperms by the process of spermatogenesis. Choose the correct one with reference to above.
1) Spermatogonia has 46 chromosomes and always undergoes meiotic cell division.
2) Primary spermatocytes divide by mitotic cell division.
3) Secondary spermatocytes have 23 chromosomes and undergo second meiotic division.
4) Spermatozoa are transformed into spermatids.

41. The diagram of the sectional view of a seminiferous tubule is shown below. Choose the correct option that includes all diploid cells.



1) A, B
2) C, D, E
3) A, B, C
4) A, B, E

42. During oogenesis, the first meiotic division begins
1) Just after birth
2) At puberty
3) Before birth
4) At menarche

43. Which of the following is released with the rupture of Graafian follicle?
1) Primary oocyte
2) Secondary oocyte
3) Oogonium
4) Second polar body

44. Read the given assertion and reason carefully and select the correct option
Assertion: The process of oogenesis is markedly different from spermatogenesis.

Reason: Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (oogonia) are formed in both foetal ovary and their number keeps on increasing even after birth.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

45. At birth, the female's ovary contains a couple of million follicles, each with a developing ovum that has begun meiosis but it remains arrested in prophase of the first meiotic division. At this stage, these cells are called

1) Primary oocyte
2) Secondary oocyte
3) Secondary follicle
4) Tertiary oocyte

46. Menstruation occurs due to abrupt decrease in the level of

1) LH
2) FSH
3) Progesterone
4) Estrogen

47. Identify the correct statement regarding menstrual cycle.

1) The luteal phase of menstrual cycle lasts 7-8 days
2) The endometrial wall is sloughed off during proliferative phase
3) During a menstrual cycle, ovulation occurs at the end of proliferative phase
4) During menstruation, the myometrium of uterine wall is shed along with the blood that does not clot.

48. Which of the following statement is applicable to the luteal phase of menstrual cycle?

1) Development of follicle takes place.
2) Regeneration of endometrium occurs through proliferation.
3) Endometrium becomes more vascular and glandular.
4) Strong contractions of myometrium are observed.

49. Which of the following hormones is not secreted during proliferative phase?

1) LH
2) FSH
3) Estrogen
4) Relaxin

Fertilization and Implantation

50. The extrusion of second polar body from the egg nucleus occurs

1) Inside ovary
2) In uterine cavity
3) In the fallopian tube after the contact of the sperm with zona pellucida and before the completion of fertilization
4) In the fallopian tube after the entry of the sperm into the cytoplasm of ovum and before the completion of fertilization

51. Read the given assertion and reason carefully and select the correct option.

Assertion: All copulations cannot lead to fertilisation and pregnancy.

Reason: Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

52. Match the following and choose the correct options

Column I		Column II	
A	Trophoblast	i	Embedding of blastocyst in the endometrium
B	Cleavage	ii	Group of cells that would differentiate as embryo
C	Inner cell mass	iii	Outer layer of blastocyst attached to the endometrium
D	Implantation	iv	Mitotic divisions of zygote

Options:

1) A-ii, B-i, C-iii, D-iv
2) A-iii, B-iv, C-ii, D-i
3) A-iii, B-i, C-ii, D-iv
4) A-ii, B-iv, C-iii, D-i

53. What happens during fertilization in humans after many sperms reach close to the ovum?

1) Only two sperms nearest to the ovum penetrate zona pellucida
2) Cells of corona radiata trap all sperms except one
3) Secretions of acrosome helps one sperm to enter cytoplasm of ovum through zona pellucida
4) All sperms except one nearest to the ovum lose their tails

54. Which of the following is not formed prior to the entry of sperm?

1) Oogonia
2) Primary oocyte
3) Secondary oocyte
4) Second polar body

55. Implantation of embryo with the uterine lining soon leads to

1) Formation of umbilical cord
2) Rise in progesterone levels
3) Contraction of myometrium
4) Fetal ejection reflex

56. Acrosomal reaction of the sperm occurs due to

1) Its contact with zona pellucida of the ova
2) Reactions within the uterine environment of the female
3) Reactions within the epididymal environment of the male
4) Androgens produced in the uterus

57. Read the given statements and choose the correct option.

Statement I: The embryo with 8 to 16 blastomeres is called a morula.

Statement II: The blastomeres in the blastocyst are arranged into an outer layer called trophoblast and an inner group of cells attached to trophoblast called the inner cell mass.

1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

58. Which of the following embryonic stages attaches to the endometrial lining of uterus for further development?

1) Gastrula

2) Morula
3) Blastocyst
4) Zygote

59. Refer to the given assertion and reason carefully and select the correct option.
Assertion: After fusion of the male and female gametes, the zygote would carry either XX or XY depending on whether the sperm carrying X or Y fertilised the ovum.
Reason: 50 per cent of sperms carry the X chromosome while the other 50 percent carry the Y chromosome.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

Pregnancy and Embryonic Development

60. Which of the following is a correct sequence of embryonic development?
1) Zygote - Blastula - Morula – Gastrula
2) Zygote - Morula - Blastula – Gastrula
3) Morula - Zygote - Blastula – Gastrula
4) Zygote - Gastrula - Blastula - Morula

61. During which month of pregnancy, the first fetal movements and appearance of hair on the head are observed?
1) After first month
2) During third month
3) During fifth month
4) At the end of second trimester

62. Which of the following contains stem cells?
1) Colostrum
2) Inner cells mass
3) Decidua capsularis
4) Uterine endometrium

63. The separation of eyelids and the formation of eyelashes are observed in the developing fetus in/by
1) Fifth month
2) First trimester
3) End of 24 weeks
4) End of 12 weeks

64. Reed the given assertion and reason carefully and select the correct option.
Assertion: During pregnancy, the levels of hormones like estrogens, progestrogen, cortisol, prolactin, etc., are increased several folds in the maternal blood.
Reason: Increased production of these hormones. is not essentially required for supporting the foetal growth.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

65. Read the given statements and choose the correct option.
Statement I: The placenta acts as an endocrine tissue.

Statement II: The placenta is connected to the embryo through an umbilical cord, which helps in the transport of substances to and from the embryo.

1) Both statement I and statement II are true.
2) Both statement I and statement II are false.
3) Statement I is true and statement II is false.
4) Statement I is false and statement II is true.

66. During pregnancy, the levels of which hormone(s) increases several folds in the maternal blood?

1) Estrogens and progestogens
2) Cortisol and prolactin
3) Estrogens, progestogens, and thyroxine
4) All of the above

67. The first sign of growing foetus may be noticed by

1) Listening to the heart sound carefully through the sphygmomanometer
2) Appearance of hair on the head
3) Listening to the heart sound carefully through the stethoscope
4) Its movement

68. Reed the given assertion and reason carefully and select the correct option

Assertion: The hormones hCG, hPL and relaxin are produced in women only during pregnancy.

Reason; The hCG, hPL and relaxin are secreted by placenta only.

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

69. Read the following statements and choose the incorrect option.

1) Mesoderm appears between the ectoderm and the endoderm.
2) The inner cell mass contains certain cells called stem cells.
3) By the end of 8 weeks (second trimester), most of the major organ systems are formed.
4) Estrogens and progestogens are also secreted by placenta.

70. Increased production of hormones during pregnancy is essential for

1) Supporting the foetal growth
2) Maintenance of pregnancy
3) Metabolic changes in the mother
4) All of the above

71. The chorionic villi and uterine tissue interdigitated with each other and jointly form

1) Placenta
2) Umbilical cord
3) Embryo
4) Stem cells

Parturition and Lactation

72. Find the wrong statement.

1) The first milk produced after childbirth is called colostrum.
2) Colostrum is rich in IgD.
3) Mother provides natural passive immunity to newborn through milk in initial few days after birth.
4) None of the above

73. In females, the role of oxytocin is

1) Milk production
2) Milk ejection
3) Strong uterine contractions during parturition
4) Both (2) and (3)

74. Fetal ejection reflex in human females is induced by
1) Release of oxytocin from pituitary gland
2) Pressure exerted by amniotic fluid
3) Fully developed fetus only
4) Fully developed fetus and placenta

75. Which of the following hormone is injected to induce labor?
1) LH
2) FSH
3) Progesterone
4) Oxytocin

76. Read the given assertion and reason carefully and select the correct option.
Assertion: Breast feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby.
Reason: The milk produced during the initial few days of lactation is called colostrum which contains several antibodies absolutely essential to develop resistance for the new born babies.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

77. Match the columns.

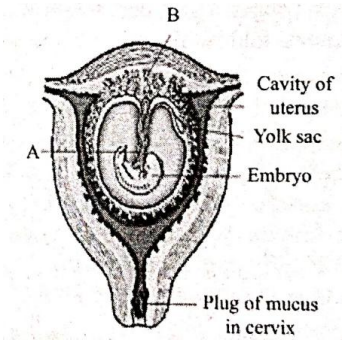
Column I		Column II	
a	Oxytocin	p	Stimulates ovulation
b	Prolactin	q	Implantation and maintenance of pregnancy
c	Luteinising hormone	r	Lactation after child birth
d	Progesterone	s	Uterine contraction during labour
		t	Reabsorption of water by nephrons

	a	b	c	d
1)	s	r	p	q
2)	s	r	p	s
3)	s	q	r	t
4)	t	p	s	r

78. The fetal ejection reflex in humans triggers the release of
1) Oxytocin from fetal pituitary
2) Human chorionic gonadotropin (hCG) from placenta
3) Human placental lactogen (hPL) from placenta
4) Oxytocin from maternal pituitary

79. The average duration of human pregnancy is about 9 months which is called the
1) Incubation period
2) Gestation period
3) Latent period
4) Parturition

80. Choose the correct option with respect to the labels A and B in the diagram given below.



1) Label A is placental villi
2) Label B is the decidual capsularis.
3) Label B is the myometrium of uterus.
4) Label A is the umbilical cord

81. Expulsion of baby out of the uterus is called

1) Parturition
2) Lactation
3) Gestation
4) Puberty

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
3	2	2	2	3	2	3	1	2	2	2	4	3	2	2
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
3	2	3	1	3	2	4	1	1	4	2	2	4	3	4
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
1	2	4	4	4	3	2	2	4	3	1	3	2	3	1
46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
3	3	3	4	4	1	2	3	4	2	1	1	3	1	2
61	62	63	64	65	66	67	68	69	70	71	72	73	74	75
3	2	3	3	1	4	3	3	3	4	1	2	4	4	4
76	77	78	79	80	81									
1	1	4	2	4	1									